NEXA

TECHNOLOGY FOR A BRIGHTER TOMORROW

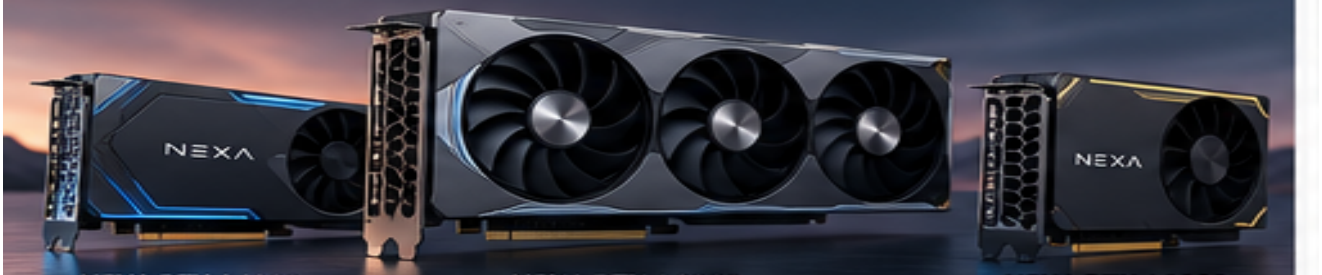
TECHNICAL GUIDE

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NexaGPU Series

Technical Guide

High-Performance GPUs for AI, Creators and Professionals



- Built for AI and ML Workloads
- High-VRAM Configurations
- Advanced Cooling Design
- Optimized for Creators and Professionals

NEXA RTX A4000	NEXA RTX A6000	NEXA RTX A2000
12 GB GDDR6X	24 GB GDDR6X	8 GB GDDR6

1. Product Lineup

Model	AI Performance (TOPS)	CUDA Cores	Memory	Memory Type	Target Use Cases
NEXA RTX A2000	125	3,072	8.GB	GDDR6	Entry-level AI, Development
NEXA RTX A4000	300	6,144	12 GB	GDDR6X	AI/ML, Content Creation
NEXA RTX A6000	600	10,752	24 GB	GDDR6X	Large AI Models, Professional Workloads

2. Key Specifications

Specification	RTX A2000	RTX A6000	RTX A6000
GPU Architecture	Nexa Lovelace	Nexa Lovelace	Nexa Lovelace
CUDA Cores	3,072	6,144	10,752
AI Performance (INT8)	125 TOPS	300 TOPS	600 TOPS
Memory	8 GB	12 GB	24 GB
Memory Type	GDDR6	GDDR6X	GDDR6X
Memory Bandwidth	256 GB/s	504 GB/s	960 GB/s
TDP	150 W	250 W	350 W
Display Outputs	3 × DisplayPort 1.4, 1 × HDMI 2.1		
Recommended PSU	550 W	650 W	850 W
Form Factor	Dual Slot	Dual Slot	Triple Slot

3. Software& Driver Support

Platform	Support
Windows 11 (64-bit)	Nexa GPU Driver (Game Ready / Studio)
Linux (Ubuntu, RHEL)	Nexa Official Driver / CUDA Toolkit
AI Frameworks	PyTorch, TensorFlow, ONNX, CUDA, cuDNN
Developer Tools	NVIDIA CUDA Toolkit, Nsight, Docker, WSL2

1. Product Lineup

The NexaGPU Series is a fictional family of discrete graphics accelerators designed for the NexaStore educational environment. The lineup demonstrates how GPU memory, compute throughput, power and cooling influence AI, creator and professional workloads.

Model	VRAM	Typical Board Power	Target workload
NexaGPU A2000	8 GB GDDR6	150 W	Entry AI development, embeddings, light computer vision and creator workflows
NexaGPU A4000	12 GB GDDR6X	250 W	Local inference, AI application development, content creation and GPU-accelerated engineering
NexaGPU A6000	24 GB GDDR6X	350 W	Larger local models, advanced computer vision, heavier creator work and professional AI experimentation

2. Architecture & Capabilities

Capability	A2000	A4000	A6000
AI acceleration	Supported	Enhanced	High
Ray tracing	Supported	Supported	Supported
Display outputs	3 x DP + 1 x HDMI	3 x DP + 1 x HDMI	3 x DP + 1 x HDMI
Recommended system memory	32 GB	64 GB	64-128 GB
Recommended PSU	550 W	650 W	850 W
Form factor	Dual slot	Dual slot	Triple slot

3. AI Workload Guidance

Workload	Primary constraint	Guidance
Embeddings / RAG development	VRAM and batch size	All three models are suitable; higher VRAM enables larger batches and concurrent workloads.
7B-class local inference	VRAM, quantization, context	A4000 and A6000 provide more headroom; A2000 may require more aggressive quantization.
14B-class local inference	VRAM and system RAM	A6000 is preferred for local experimentation; actual feasibility varies by model and runtime.
Computer vision	GPU compute and VRAM	A4000/A6000 are recommended for larger image batches and model experimentation.
Fine-tuning experiments	VRAM and optimizer memory	Use smaller models, adapters or parameter-efficient methods; cloud/data-center GPUs may still be preferable.

4. Software & Driver Support

Platform	Notes
Windows 11 64-bit	Use current Nexa-qualified GPU driver package.
Linux	Use a supported distribution and compatible GPU compute stack.
Containers	GPU access requires a compatible host driver and container runtime configuration.
.NET / Python AI apps	Applications may use supported model runtimes and provider SDKs; GPU access depends on runtime compatibility.

5. Installation Notes

Before installation, verify chassis clearance, power-supply capacity, auxiliary power connectors and airflow. Disconnect AC power before opening the chassis. After installation, update the GPU driver and validate temperatures under sustained workloads.

Educational notice

NexaGPU products are fictional and exist only for the NexaStore course case study. Specifications are intentionally controlled for RAG, recommendation and agent demonstrations.